To get started with Git, GitHub, and Visual Studio Code (VS Code) using the command line (cmd/terminal), here is a step-by-step guide designed for first-time users:

Step 1: Install Git and Check Setup

If you do not have Git installed on your computer yet, download and install it from git-scm.com.

Once installed, open your terminal (Command Prompt on Windows or Terminal on Mac/Linux) and configure your identity so GitHub knows who is making changes:

- Type git config --global user.name "Your Full Name" and press Enter.
- Type git config --global user.email "your.email@example.com" and press Enter.

Step 2: Create a Repository on GitHub

1. Navigate to GitHub.com and log into your account.
2. Click the New button or the + icon in the top right to create a new repository.
3. Name your repository (e.g., capstone-app), choose public or private, check the box to "Add a README file", and click Create repository.

Step 3: Open Your Project in VS Code

1. Open Visual Studio Code.
2. Open your project folder by clicking File > Open Folder... and selecting or creating your project folder (e.g., capstone-app).
3. Open the integrated terminal inside VS Code by pressing Ctrl + ~ (or Cmd + ~ on Mac), or by clicking Terminal > New Terminal from the top menu.

Step 4: Initialize Git Locally and Connect to GitHub

Run the following commands sequentially in your VS Code terminal to link your local folder to your remote GitHub repository:

- Initialize local tracking: git init (Initializes an empty Git repository in your folder).
- Check the status of your files: git status (Shows any untracked or modified files).
- Stage all your files: git add . (Packages all current changes to be saved).
- Commit your snapshot: git commit -m "Initial project setup" (Permanently records your staged changes with a descriptive message).
- Ensure your default branch is named main: git branch -M main.

- Connect your local folder to GitHub: git remote add origin https://github.com/your-username/capstone-app.git (Replace with your actual repository URL).
- Push your code to the cloud: git push -u origin main (Uploads your commits to GitHub).

Troubleshooting Tip: If you encounter an error stating that Updates.were.rejected. because.the.remote.contains.work.that.you.do.not.have (which often happens if a README was automatically created on GitHub), run git pull origin main --allow-unrelated-histories to merge them, then run git push -u origin main again.

Step 5: The Everyday Git Habit

Moving forward, practice the professional save cycle every time you write or update code:

1. Check status: git status
2. Stage changes: git add .
3. Commit changes: git commit -m "Describe what you changed here"
4. Push to GitHub: git push